MIL-G-51066C
9 April 1970
SUPERSEDING
MIL-G-0051066B(MU)
31 March 1966
MIL-G-51066A
25 June 1963

MILITARY SPECIFICATION

GENERATOR, SMOKE, MECHANICAL

PULSE JET, M3A3

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers a smoke generator for use in producing smoke screens.

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

P-D-680	-	Dry Cleaning Solvent.
FF-N-105	-	Nails, Wire, Brads, and Staples.
NN-P-515	-	Plywood, Container Grade.
QQ-S-781	-	Steel, Strapping, Flat.
PPP-B-566	-	Boxes, Folding, Paperboard.
PPP-B-621	-	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-640	-	Box, Fiberboard, Corrugated, Triple-Wall.
PPP-T-60	-	Tape: Pressure-Sensitive Adhesive, Waterproof, for Packaging.
PPP-T-97	-	Tape: Pressure-Sensitive Adhesive, Filament Reinforced.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:48Z
Check the source to verify that this is the current version before use.

MILITARY

MIL-P-116 - Preservation, Methods of.
MIL-R-196 - Repair Parts for Internal Combustion Engines, Packaging of.
MIL-T-704 - Treatment and Painting of Materiel.
MIL-F-12070 - Fog Oil.
MIL-V-13811 - Varnish, Waterproofing, Electrical Ignition.
MIL-H-15424 - Hand Tools; Packaging of.
MIL-L-25567 - Liquid, Leak Detector.
MIL-B-26195 - Boxes, Wood-Cleated, Skidded, Load-Bearing Base.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-731 - Quality of Wood Members for Containers and Pallets.

DRAWINGS

US ARMY MUNITIONS COMMAND

EDGEWOOD ARSENAL

DL31-15-1259 - Generator, Smoke, Mechanical, Pulse Jet, M3A3.
IDL 31-15-1259 - Generator, Smoke, Mechanical, Pulse Jet, M3A3 Inspection Data List.

PUBLICATIONS

DEPARTMENT OF THE ARMY

TM3-1040-202-12 - Generator, Smoke, Mechanical, Pulse Jet, M3A3.

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:48Z
Check the source to verify that this is the current version before use.

UNIFORM CLASSIFICATION COMMITTEE

Uniform Reight Classification

(Application for copies of these ratings, rules and regulations should be addressed to the Uniform Classification Committee, 202 Union Station, 516 West Jackson Boulevard, Chicago, Illinois 60606.)

3. REQUIREMENTS

3.1 Materials and components.

3.1.1 Materials. All materials cited on Drawing DL31-15-1259 and on the subsidiary drawings shall conform to the specifications listed thereon or to the specific characteristics set forth on the drawings.

3.1.2 Components. All components of the generator shall conform to the specifications and drawings listed on Drawing DL31-15-1259 and subsidiary drawings.

3.2 Assembly. The generator shall be assembled as shown on Drawing E31-15-1259.

3.3 Leakage.

3.3.1 Fuel tank (Dwg D31-15-1021). The fuel tank shall not leak air when subjected to an internal pressure of 3 ± 1/4 pound per square inch gage (psig) for a period of 30 + 10 seconds. The joint between the overflow well and fuel tank shall not leak water during a period of 30 + 10 seconds when tested in accordance with 4.4.4.1(a).

3.3.2 Engine tube assembly (Dwg E31-15-1269 or E31-15-1271). The engine tube shall not leak air when subjected to an internal pressure of 30 ± 1 psig for a period of 2 + 1 minute when tested in accordance with 4.4.4.1(b).

3.3.3 Engine tube housing assembly (Dwg D31-15-1024). The engine tube housing shall not leak air when subjected to an internal pressure of 10 + 1 psig for a period of 60 + 10 seconds when tested in accordance with 4.4.4.1(c).

3.3.4 Cable shielded ignition (Dwg C31-15-1296). The cable with sleeves removed shall not leak air when subjected to a pressure of 4 + 1 psig for a period of 15 + 5 seconds when tested in accordance with 4.4.4.1(d).

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:48Z
Check the source to verify that this is the current version before use.

3.4 Functioning.

3.4.1 Flowjector (Dwg C31-15-1101). The flowjector shall raise a column of water a minimum height of 9 inches when pressurized to 30 + 1 psig when tested in accordance with 4.4.4.2(a).

3.4.2 Float bowl assembly (Dwg C31-15-1039). The flow of gasoline shall cease when the bowl float is in the shut-off position. The gasoline shall flow when the plunger is depressed, and cease flowing when the plunger is released when tested in accordance with 4.4.4.2(b).

3.4.3 Generator assembly (Dwg E31-15-1259).

3.4.3.1 Functioning. The generator assembly shall be functional for 30 + 5 minutes. It shall produce smoke and shall consume fog oil at a minimum rate of 40 gallons per hour (gph) when the oil metering valve is fully opened. During production of smoke, fog oil shall flow freely through the exhaust hose. There shall be no evidence of liquid leakage when tested in accordance with 4.4.4.2(c).

3.4.3.2 Endurance. The generator shall be operated continuously for a period of 4 + 1 hour. Minor repairs will be permitted and include as part of the test time providing the time does not exceed 15 minutes. Replacement of parts shall be limited to generator "On board" repair parts. Fog oil consumption in accordance with paragraph 3.4.3.1 shall be verified at approximately two hour intervals. Test shall be in accordance with 4.4.4.2(d).

3.4.5 Magneto air pump assembly (Dwg D31-15-1052).

3.4.5.1 Air. The pump shall deliver a minimum of 30 psig within 30 seconds when the pump is operated at a speed of 60 ± 5 forward strokes per minute. Each stroke shall be 10 ± 2 inches as measured between the magnet air pump clamp and handle lock nut. Test shall be in accordance with 4.4.4.2(e)(1).

3.4.5.2 Voltage. The magneto shall deliver voltage sufficient to permit an electrical discharge across a 1/8-inch minimum air gap when the pump is operated at the speed and length of stoke specified in 3.4.5.1. This operation shall continue for 15 to 20 seconds and a minimum of 20 seconds and a minimum of 20 repetitive electrical discharges across the electrodes shall be observed when tested in accordance with 4.4.4.2(e)(2).

3.5 Preproduction. Prior to the start of production, a preproduction lot of generators shall be produced in accordance with this specification for examination and test (see 4.3).

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:48Z
Check the source to verify that this is the current version before use.

3.6 Workmanship. The generator and components shall be free from damage which could affect functioning.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.

4.1.1 Supplier's responsibility. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to specified requirements.

4.1.2 Objective evidence. The supplier shall provide objective evidence acceptable to the contracting officer that the requirements of 3.1 and section 5 for which specific inspection has not been provided in this specification have been satisfied.

4.2 Classification of inspections.

(a) Preproduction inspection (see 4.3)
(b) Quality conformance inspection (see 4.4)

4.3 Preproduction.

4.3.1 Lot. Prior to the start of regular production, two generators shall be produced using the same methods, materials, and equipment that will be used during regular production.

4.3.2 Examination and test. The generators shall be examined and tested for all the requirements of this specification.

4.3.3 Acceptance/rejection criteria. The generators shall meet all the requirements of this specification to be acceptable. The supplier shall obtain written approval from the contracting officer prior to proceeding with regular production.

4.4 Quality conformance inspection.

4.4.1 Lotting. A lot of components or smoke generators shall consist of like components or smoke generators produced by one manufacturer, at one plant, from the same materials and under essentially the same manufacturing conditions.

Source: https://assist.dla.mil -- Downloaded: 202640-22T20:48Z
Check the source to verify that this is the current version before use.

4.4.2 Sampling.

4.4.2.1 For examination and nondestructive test. Sampling shall be conducted in accordance with MIL-STD-105.

4.4.2.2 For endurance. Sampling shall be conducted in accordance with MIL-STD-105, level S-4, using an AQL of 2.5 percent.

4.4.3 Inspection procedure.

4.4.3.1 For examination and nondestructive test. Sample components and smoke generator assemblies, including levels A and B preparation for delivery shall be examined in accordance with the classification of defects and with MIL-STD-105. For endurance test use an AQL of 2.5 percent defective.

4.4.3.2 For critical defects. Each item in the lot shall be tested for its critical characteristic(s) listed in the classification of defects.

4.4.3.3 Classification of defects. (See inspection data list, IDL 31-15-1259, for inspection equipment required to perform the examinations specified in this section.)

(a) Tank, fuel assembly (Dwg D31-15-1021).

Categories	Defects	Acceptance standards
Critical:		
1	Leakage	4.4.4.1(a)
Major:	AQL 1.0 percent defective	
101	Component missing or incorrectly assembled	
102	Cutlet box thread (1/4"-18) incorrect (Dwg A31-15-1022)	
103	Cutlet fitting thread (1/4-18 NPT) incorrect (Dwg A31-15-802)	
104	Drain tube plugged	
105	Workmanship (3.6)	
106	Welding, riveting, brazing, or soldering incorrect	
107	Tank length, height, or width incorrect	
Minor:	AQL 2.5 percent defective	
201	Protective finish missing	

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:48Z
Check the source to verify that this is the current version before use.

4.4.3.3 Classification of defects (continued).

(b) Pad, insulation assembly (Dwg D31-15-1279).

Categories		Defects
Critical:		None defined
Major:	AQL 1.0 percent defective	
101		Component missing or incorrectly assembled
102		Breather hole missing
103		Welding incorrect
104		Workmanship (3.6)

(c) Blanket, assembly (Dwg D31-15-891).

Categories		Defects
Critical:		None defined
Major:	AQL 1.0 percent defective	
101		Component missing or incorrectly assembled
102		Shielded breather missing
103		Welding incorrect
104		Workmanship (3.6)

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:48Z
Check the source to verify that this is the current version before use.

4.4.3.3 Classification of defects (continued).

(d) Tube, flanged, engine, assembly; or tube assembly, engine, assembly (Dwg E31-15-1271 or Dwg E31-15-1269).

Categories	Defects	Acceptance standards
Critical:		
*1	Engine tube assembly leakage	4.4.4.1(b)
Major:	AQL 1.0 percent defective	
101	Component missing or incorrectly assembled	
102	Pressurizing line boss thread incorrect (Dwg A31-15-1158)	
103	Spark plug boss thread incorrect (Dwg D31-15-1274)	
104	Combustion chamber ring inside diameter incorrect (Dwg B31-15-867)	
105	Oil injection nozzle thread (7/8-14) incorrect (Dwg B31-15-1277)	
106	Front plate mounting hole locations or diameters incorrect (Dwg D31-15-1273)	
107	Welding incorrect	
108	Workmanship (3.6)	
109	Combustion chamber adapter dimension incorrect (0.063) (Dwg B31-15-865)	
110	Combustion chamber adapter cutouts dimension or location incorrect (Dwg B31-15-865)	
111	Combustion chamber ring machine finish 63 incorrect (Dwg D31-15-1274)	

*For prime procurement only.

(e) Head, engine, assembly (Dwg C31-15-1038).

Categories	Defects
Critical:	None defined
Major:	AQL 1.0 percent defective
101	Component missing or incorrectly assembled
102	Petals not coverin g valve holes
103	Workmanship (3.6)

Source: https://assist.dla.mil -- Downloaded: 202640-22T20:48Z
Check the source to verify that this is the currentversion before use.

4.4.3.3 Classification of defects (continued).

(f) Adapter, head (Dwg C31-15-866).

Categories	Defects
Critical:	None defined
Major:	AQL 1.0 percent defective
101	Thread (3-1/8"-12) incorrect
102	Ear dimensions or location incorrect
103	Holes missing
104	Machine finish 63 incorrect
105	Workmanship (3.6)
Minor:	AQL 2.5 percent defective
201	Outside diameter under ears incorrect

(g) Head, engine (Dwg C31-15-1037).

Categories	Defects
Critical:	None defined
Major:	AQL 1.0 percent defective
101	Thread (5/16"-18) incorrect
102	Thread (1/4"-20) incorrect
103	Thread (3-1/8"-12) incorrect
104	Venturi throat diameter incorrect
105	Depth (1.437) incorrect
106	Threaded shoulder diameter incorrect
107	Porthole diameters incorrect
108	Porthole locations incorrect
109	Keyway dimensions, radius, or location incorrect
110	Face not flat
111	Porthole angle (30-1/2°) incorrect
112	Venturi angle (12-1/2°) incorrect
113	Eccentricity of 5/16" tapped hole excessive
114	Machine finish incorrect
115	Workmanship (3.6)

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:48Z
Check the source to verify that this is the current version before use.

4.4.3.3 Classification of defects (continued).

(g) Head, engine (Dwg C31-15-1037)(continued).

Categories		Defects
Critical:		None defined
Minor:	AQL	2.5 percent defective
201		Overall length incorrect
202		Counterbore diameter incorrect
203		Fin width or spacing incorrect
204		Minimum of 5/16"-18 thread incorrect
205		Venturi radius incorrect
206		Protective finish incorrect or missing
207		Width across flats incorrect

(h) Valve, engine (Dwg C31-15-1036).

Categories		Defects
Critical:		None defined
Major:	AQL	1.0 percent defective
101		Profile incorrect
102		Hole diameter or location incorrect
103		Slot dimensions or location incorrect
104		Thickness incorrect
105		Locations of petals with respect to slot center lines incorrect
106		Hardness incorrect
107		Workmanship (3.6)

Source: https://assist.dla.mil -- Downloaded: 202610-22T20:48Z
Check the source to verify that this is the currentversion before use.

4.4.3.3 Classification of defects (continued).

(i) Flowjector (Dwg C31-15-1101).

Categories	Defects	Acceptance standards
Critical:	Function failure	4.4.4.2(a)
Major:	AQL 1.0 percent defective	
101	Thread (5/16"-32) incorrect	
102	Thread (1/4"-20) incorrect	
103	Thread (1/4"-32) incorrect	
104	Seating shoulder diameter (0.300") incorrect	
105	Air tube location in relation to gasoline cutlet hole incorrect	
106	Gasoline-outlet-holes diameters (0.041") or location incorrect	
107	Burr on 0.041 diameter holes	
108	Diameter (0.180") incorrect	
109	Workmanship (3.6)	
Minor:	AQL 2.5 percent defective	
201	Minimum depth of 1/4-32 thread incorrect	

(j) Jet, metering (Dwg B31-15-1100).

Categories	Defects
Critical:	None defined
Major:	AQL 1.0 percent defective
101	Thread incorrect
102	Thread location incorrect
103	Marking incorrect or missing
104	Hole diameter (0.059") incorrect
105	End diameter (0.190") incorrect
106	Hole diameter (0.081) incorrect
107	Workmanship (3.6)

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:48Z
Check the source to verify that this is the current version before use.

4.4.3.3 Classification of defects (continued).

(k) Case, tool kit, assembly (Dwg D31-15-793).

Categories	Defects
Critical:	None defined
Major:	AQL 1.0 percent defective
101	Inability of case to properly contain *tools in closed position*

(l) Generator, smoke, mechanical, pulse jet, M3A3 (Dwg E31-15-1259).

Categories	Defects	Acceptance standards
Critical:		
1	Engine-tube-assembly leakage	4.4.4.1(b)
2	Engine-tube-housing-assembly leakage	4.4.4.1(c)
3	Functioning	4.4.4.2(c)
Major:	AQL 1.0 percent defective	
101	Component missing or incorrectly assembled	
102	Welding or soldering incorrect	
103	Thread compound missing or incorrect	
104	Identification plate missing or incorrectly marked	
105	Workmanship (3.6)	
Minor:	AQL 2.5 percent defective	
201	Tool box hardware missing	
202	Tool box drain hole missing	
203	Carrying handle not free	
204	Protective finish missing or incorrect	

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:48Z
Check the source to verify that this is the current version before use.

4.4.3.3 Classification of defects (continued)

(m) Packing, smoke generator, assembly (section 5).

Categories		Defects
Critical:		None defined
Major:	AQL	1.0 percent defective
101		Contents, preservation missing or incorrect
102		Contents missing or incorrect
103		Contents incorrectly located or secured
104		Strap missing, loose, or incorrect
105		Marking incorrect or missing
106		Workmanship (3.6)
Minor:	AQL	2.5 percent defective
201		Tape missing, inadequate, or incorrect
202		Loose nails

4.4.4 Tests. Tests shall be conducted as follows: (See inspection data list, IDL 31-15-1259 for inspection equipment required to perform the tests specified in this section.)

4.4.4.1 Leakage.

(a) Fuel tank assembly (air test). Subject the tank to the specified pressure, submerge in water and maintain the pressure for the required period of time. If no leakage is observed, plug the discharge outlet of the overflow well and fill the well to a depth of 1/2-inch and hold for the required period of time. The filling cap shall be in place on the fuel tank during this test.

(b) Engine tube assembly. Bolt the engine tube assembly with the engine head assembly in place to the inner shell with the gasket in place. Plug the oil injection and pressurizing holes and run the engine for periods of 1 minute, 45 seconds, and 30 seconds with a cooling period of 5 minutes between each period of operation. After the last cooling period, remove the engine tube assembly from the shell. Subject the assembly to the specified pressure, submerge in water and maintain the pressure for the required period of time.

Source: https://assist.dla.mil -- Downloaded: 202610-22T20:48Z
Check the source to verify that this is the currentversion before use.

(c) Engine tube housing assembly. Assemble the engine tube assembly (Dwg E31-15-1269 or Dwg E31-15-1271), the outer discharge baffle (Dwg D31-15-1125), the backup ring (Dwg D31-15-1025) and the gasket (Dwg C31-15-1150) with the engine tube housing assembly (Dwg D31-15-1024) in accordance with Drawing E31-15-1278. Mount the assembly so that all exit air openings are closed. Subject the assembly to the specified pressure for the required period of time. Apply a leak detector solution meeting the requirements of MIL-L-25567 to all joints and connections and observe for leakage.

(d) Cable shielded ignition. Subject the cable to the specified pressure, submerge in water and maintain the pressure for the required period of time.

4.4.4.2 Functioning.

(a) Flowjector. With a test device attached to the metering jet (Drawing B31-15-1100) assembled to the flowjector, introduce the specified air pressure through the air inlet of the flowjector.

(b) Float bowl. Connect the float bowl to a source of gasoline raised 1 foot above the top surface of the float bow. Close the needle valve for a period of 60 + 5 seconds by allowing a flow of gasoline to raise the float. Slowly depress the spring-loaded plunger on top of the float bowl 5 times.

(c) Generator assembly. Operate the generator in accordance with TM3-1040-202-12 using fog oil conforming to type SFG2 of MIL-F-12070. Fully open the oil metering valve and place both the exhaust and intake hoses in a container of fog oil. After the generator has been producing smoke, determine the rate of fog oil consumption by measuring the difference in volume or weight of oil in the container over a known period of no less than fifteen minutes. Repeat the test using the repair engine head furnished with each generator.

(d) Generator endurance. Operate the generator in accordance with TM3-1040-202-12 for the required period of time.

(e) Magneto air pump assembly.

(1) Air. Connect a hose of suitable length to the 1/8-27NPT air discharge hole and to a receiving air tank at 0 psig internal air pressure. The tank shall have a capacity of 100 ± 10 cubic inches and be equipped with a suitable pressure gage, a bleed valve, and a valve to check reverse flow from the intake opening of the air tank. Operate the pump at the specified speed and stroke.

Source: https://assist.dla.mil -- Downloaded: 202640-22T20:48Z
Check the source to verify that this is the current version before use.

(2) Voltage. Connect a cable of suitable length to the magneto electrical outlet and to a 14-millimeter internal resistor spark plug, with the electrode gas set at 1/8-inch minimum. Operate the pump for the required period of time.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. All packaging and packing shall be of the minimum cube required to contain the items.

5.1.1 Cleaning and drying. After completion of test runs, if any, all surfaces to which a preservative or coating is to be applied shall be cleaned and dried in conformance with MIL-P-116, using cleaning method C-1 and any of the drying methods listed therein.

5.1.2 Repainting. Damaged portions of painted surfaces shall be treated and painted in accordance with MIL-T-704, type A, except that the color of the finish coat shall conform to the requirements shown on the applicable drawings.

5.1.3 Level A. The level of packaging shall be as specified in the contract or order (see 6.2).

5.1.3.1 Preservatives and methods of preservation. Preservatives and their application as well as the methods of preservation shall be in conformance with MIL-P-116. Preservatives shall not be applied to painted surfaces, and provision shall be made to prevent such parts from coming in contact with preservatives on adjacent metal surfaces.

5.1.3.2 Threaded and other unpainted exposed metallic surfaces. Unless otherwise specified hereinafter, threaded and other unpainted exposed metallic surfaces, including such surfaces exposed by disassembly, shall be coated with type P-1 preservative compound.

5.1.3.3 Prime unit.

(a) Preservative run-in. The strainer shall be removed from the fog oil intake line, cleaned, and replaced. The fog oil intake and overflow lines shall then be placed in a drum containing type F-10, grade 2 preservative oil. The cap shall be removed from the fog oil injection nozzle, and the smoke generator tilted to prevent oil from collecting in the engine tube assembly. A 3 to 15-psig air supply shall be connected to the motor, and the pump operated until the preservative oil flows freely from the nozzle drain. The air supply shall be closed off and the bleed valve opened. The excess oil shall be drained from the injection nozzle and from the oil intake and overflow lines. The air pres-

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:48Z
Check the source to verify that this is the current version before use.

surizing line shall be disconnected from the oil pump air motor, and the engine head assembly removed. The same grade of preservative oil shall then be fogged through the air pressurizing line at the point of disconnection from the motor until oil appears in the combustion chamber. The excess oil shall then be drained from the line, and the line reconnected to the motor. With the bleed valve open, the drain plugs shall be removed and the excess oil drained from the motor housing. The bleed valve shall then be closed. The threaded surfaces of drain plugs and drain plug openings shall be preserved as specified in 5.1.3.2 and the plugs packaged in accordance with method IC-3, securing them in a conspicuous location near the oil pump air motor and attaching a tag indicating that they are to be installed prior to operating the motor.

(b) Engine head assembly. The removed engine head assembly shall be flushed clean with petroleum solvent conforming to P-D-680. All solvent shall be drained from the engine head and the assembly completely dried, using drying method D-1. The engine head assembly shall then be flushed with type P-9 preservative oil to thoroughly cover all internal surfaces. The excess preservative oil shall be drained and the engine head assembly reinstalled. The spark plug shall be coated with type P-10, grade 2 preservative oil and the plug replaced.

(c) Fuel tank assembly. All gasoline shall be drained from the fuel tank and lines, and the strainer removed, cleaned, and replaced. The valves, except for the float bowl shut-off valve, shall be opened and the tank completely filled with type P-10, grade 2 preservative oil. The float bowl plunger shall be actuated until the bowl overflows. The hose shall be disconnected from the fuel line; the shutoff valve and the fuel tank drain shall be opened. The valves and drain cock shall be left open. The filler cup and filler neck shall be coated with the same grade of preservative oil. The tank cap shall be secured in place with class 1, pressure-sensitive tape conforming to PPP-T-60, and the tape overcoated with varnish conforming to MIL-V-13811.

(d) Starting air group. During the following procedure, the fuel line hose and the spark plug shall not be connected: The air line shall be disconnected from the engine head assembly. The pump handle shall be with-drawn to its maximum stroke. Type P-9 preservative oil shall then be pressure-fogged trough the air line into the pump in sufficient quantity to insure that the oil has entered the pump chamber. The pump handle shall then be actuated to discharge the excess oil. After draining, the air line shall be reconnected to the engine head assembly. The teeth of the pump plunger rod shall be coated with type P-10, grade 2 preservative oil, and the plunger actuated slowly to insure that the oil penetrates to the internal geared surfaces of the magneto.

Source: https://assist.dla.mil -- Downloaded: 202640-22T20:48Z
Check the source to verify that this is the current version before use.

(e) Gages. The glass of the gage dials shall be covered with a piece of fiberboard cut to the proper size and secured in place with class 1 pressure-sensitive tape conforming to PPP-T-60.

(f) Hinges. All hinges shall be treated with type P-9 preservative oil, the hinges being worked during application to insure complete penetration. All excess oil shall be wiped off.

(g) Openings. Unless otherwise specified, all openings into the interior of the unit shall be covered with class 1 pressure-sensitive tape conforming to PPP-T-60, and the taped openings shall be overcoated with varnish conforming to MIL-V-13811.

5.1.3.4 Accessories, tools, and repair parts. Accessories, tools, and repair parts shall be preserved and packaged in accordance with the applicable requirements of MIL-R-116 and MIL-P-15424 and with the following:

(a) Engine head assembly. The spare repair engine head assembly shall be preserved as specified in 5.1.3.3.2 and packaged in accordance with method 1A-15, using a paperboard folding box conforming to PPP-B-566 as the inner container.

(b) Miscellaneous mount hardware. Miscellaneous mount hardware shall be coated with type P-9 preservative and packaged in accordance with method 1A-8 and secured as specified on Drawing B18-60-1088.

(c) Tube, flexible nozzle. The tube shall be preserved with type P-9 preservative and packaged in accordance with method CIC-1.

(d) Board lapping. This item shall be packaged in accordance with method III.

(e) Case, tool kit. Tools and accessories, preserved and packaged as specified in 5.1.3.4, shall be placed and closed in the tool kit case shown on Drawing D31-15-793. The case shall be packaged in accordance with method III.

5.1.3.5 Technical publications. Technical publications shall be packaged in accordance with method 1C-2 or 1C-3.

5.1.4 Level C. The smoke generator, including accessories, tools, repair parts and technical publications shall be preserved and packaged to provide adequate protection against corrosion, deterioration and damage from the supply source to the first receiving activity for immediate use.

Source: https://assist.dla.mil -- Downloaded: 202610-22T20:48Z
Check the source to verify that this is the current version before use.

5.2 Packing. The level of packing shall be as specified in the contract or order (see 6.2).

5.2.1 Level A. The smoke generator and its accessories, tools, repair parts, and technical publications, preserved and packaged as specified in 5.1 shall be packed as shown on Drawing B18-60-1088 and shall be enclosed in a cleated plywood superstructure conforming to class 2, style C of MIL-B-26195. Attachment of the superstructure to the base shall be in accordance with the requirements of MIL-B-26193.

5.2.2 Level B. The smoke generator and its accessories, tools, repair parts and technical publications preserved and packaged as specified in 5.1 shall be packed as specified on Drawing B18-60-1088 and shall be enclosed as shown on Drawing B18-60-1089.

5.2.3 Level C. The smoke generator and its accessories, tools, repair parts, and technical publications, preserved and packaged as specified in 5.1 shall be packed to insure carrier acceptance and safe delivery to the first domestic destination. Containers shall be in compliance with Uniform Freight Classification rules or the regulations of other common carriers applicable to the mode of transportation.

5.3 Marking. In addition to any special marking required by the contract or order, unit packages shall be marked in accordance with MIL-STD-129. Shipping containers shall be marked in accordance with Drawing D150-2-1.

6. NOTES

6.1 Ordering data Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Level of preservation, packaging and packing required.
(c) Preproduction:

(1) Time allowed for suppliers to submit samples for Government test and evaluation after award of contract.
(2) Name and address of test facility and shipping instructions when testing is performed by the Government.
(3) Time required for the Government to notify the supplier whether to proceed with production.

6.2 Repair parts packaging. When repair parts packaging and packing is required for items authorized for stocking as repair parts, the repair parts data sheets covering the applicable components of the smoke generator should be used. (Copies of repair parts packaging data sheets may be obtained from the Technical Support Directorate, Attn: SMUEA-TSE-D, US Army Edgewood Arsenal, Edgewood Arsenal, Maryland 21010.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:48Z
Check the source to verify that this is the current version before use.

6.3 The following specifications reflect the components required for the complete assembly of this item:

MIL-P-51062 - Pump, Fog Oil, Smoke Generator, M4.
MIL-M-51240(MU) - Mount, Smoke Generator, M2.

Custodian:

Army - MU

Preparing activity:

Army - MU(EA)

Project No. 1040-0076

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:48Z
Check the source to verify that this is the current version before use.

FOLD

POSTAGE AND FEES PAID

U.S.MAIL

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

Commander
Frankford Arsenal
Attn: SARFA-MDM
Philadelphia, PA 19137

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:48Z
Check the source to verify that this is the current version before use.

SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

Source: https://assist.dla.mil -- Downloaded: 202610-22T20:48Z
Check the source to verify that this is the current version before use.